

2018-Ph-H1-Q14

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

A lift accelerates upward with a force of 15,000 N. The mass of the lift is 1200 kg.

What is the acceleration?

Solution:

First, find weight:

$$W = mg = 1200 \times 9.8 = 11,760 \text{ N}$$

$$\text{Net force} = 15,000 - 11,760 = 3240 \text{ N}$$

$$\text{Then apply } a = \frac{F}{m} = \frac{3240}{1200} = 2.7 \text{ m/s}^2$$

Answer: 2.7 m/s²

Revision Tip:

If forces act in opposite directions (e.g. lift vs gravity), subtract them to get the net force before using $a = F/m$.